



IILM UNIVERSITY, GREATER NOIDA

Internship Report Presentation

Artificial Intelligence Internship

Student Name	Vaibhav Pathak
Enrollment No.	25SCS2040003695
Program & Semester	MCA, 3rd Semester (Section 2MCA1)
Organization	Codec Technologies

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About Codec Technologies

Codec Technologies is a technology training and skill-development organization that offers structured, industry-oriented internship and certification programs for students and early-career professionals in emerging domains such as Artificial Intelligence, Machine Learning, and Data Science.

The organization focuses on bridging the gap between academic coursework and industry-relevant, hands-on skills through mentor-guided, project-based learning — with a curriculum built around real implementation rather than theory alone.

codectechnologies.in

FOCUS AREA

AI, ML & Data Science Training

PROGRAM TYPE

Mentor-guided, project-based

MY PROGRAM

Artificial Intelligence Internship

DURATION

1 Month (27 Jul – 27 Aug 2026)

The Program in Numbers

1

MONTH

Full-time program duration

4

WEEKS

Structured learning modules

2

PROJECTS

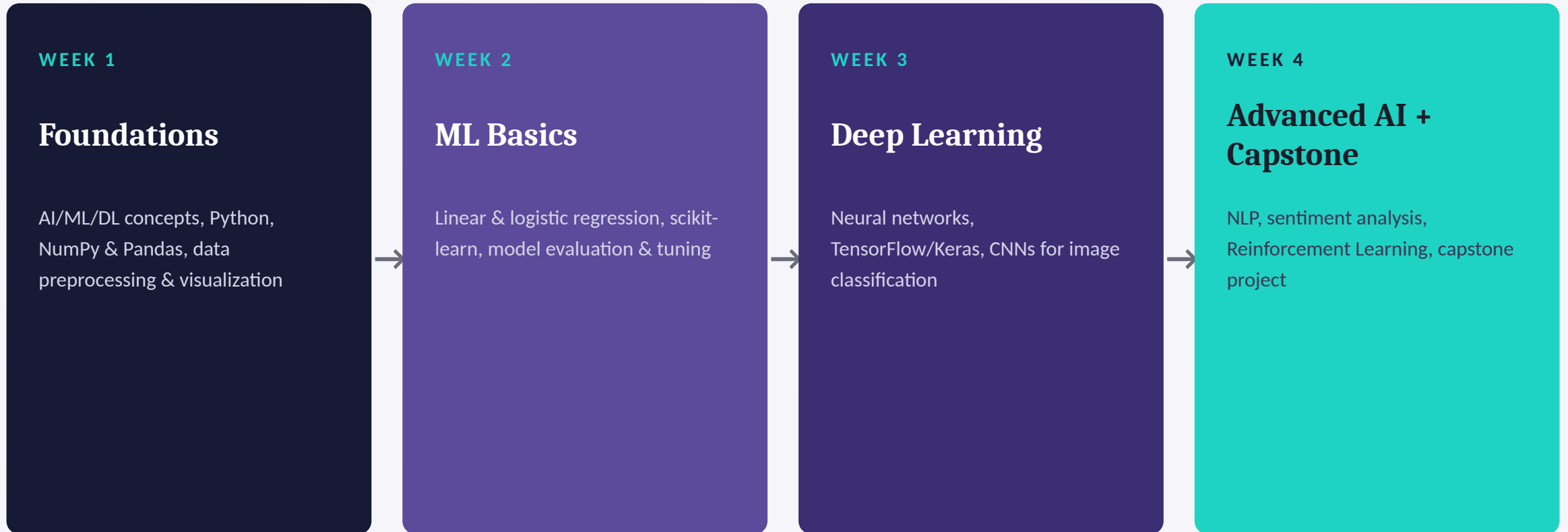
Complete AI/ML applications built

97%

ACCURACY

Achieved on the digit recognizer

A Four-Week Learning Journey



Weeks 1–2: Foundations & Machine Learning

1

Week 1: Foundations

- AI vs. Machine Learning vs. Deep Learning, and industry applications
- Python fundamentals — data structures, control flow, functions
- Data handling with NumPy & Pandas
- Data cleaning, normalization & visualization (Matplotlib, Seaborn)

2

Week 2: ML Basics

- Linear regression for continuous prediction
- Logistic regression for binary classification
- Model building with scikit-learn
- Evaluation metrics: accuracy, precision, recall, F1-score
- Train-test split, cross-validation & hyperparameter tuning

Weeks 3–4: Deep Learning & Advanced AI

3

Week 3: Deep Learning

- Neural network fundamentals — layers, weights, activation functions
- Building & training networks with TensorFlow / Keras
- Convolutional Neural Networks (CNNs) for image data
- Applied directly: Handwritten Digit Recognizer project

4

Week 4: NLP, RL & Capstone

- NLP fundamentals — text preprocessing, sentiment analysis
- Introduction to Reinforcement Learning & Q-learning
- Capstone project: selecting, building, evaluating a model
- Applied directly: Spam Message Classifier project

Handwritten Digit Recognizer

Image classification using a neural network



What was built

A neural network (MLPClassifier, two hidden layers of 64 and 32 neurons) trained to classify handwritten digits 0–9. The pipeline includes feature scaling, training, evaluation via a classification report and confusion matrix, and visualization of sample predictions.

97%

Test Accuracy

Achieved on unseen digit images

Spam Message Classifier

Text classification using Natural Language Processing



What was built

A complete NLP pipeline that converts raw text messages into TF-IDF feature vectors and classifies them as spam or ham using a Multinomial Naive Bayes model — including a command-line tool to classify new, user-supplied messages with a confidence score.

88%+

Test Accuracy

On held-out spam/ham messages

Technologies & Tools Used



Python 3

Core programming language



NumPy & Pandas

Data handling & manipulation



Matplotlib & Seaborn

Data visualization



scikit-learn

Classical ML models & evaluation



TensorFlow / Keras

Neural networks & deep learning



NLTK / TF-IDF

Text processing for NLP



Git & GitHub

Version control & submission



venv

Isolated development environment

Key Learnings & Skills Gained

1

Conceptual clarity

A solid understanding of how AI, ML, and Deep Learning relate to and differ from one another.

2

End-to-end pipelines

Ability to take a problem from raw data through preprocessing, training, evaluation, to a usable output.

3

Model evaluation

Practical fluency in accuracy, precision, recall, F1-score, and confusion matrices — not just training models blindly.

4

Two domains of AI

Hands-on experience across both vision-based (CNN/image) and text-based (NLP) classification tasks.

5

Professional practice

Documenting, packaging, and version-controlling projects the way they'd be delivered in industry.

6

Stronger foundation

A base to build on for advanced AI/ML coursework and roles, alongside existing full-stack skills.

CONCLUSION

A Strong Step Into Applied AI

This one-month Artificial Intelligence Internship with Codec Technologies took my understanding of AI from theory to practice — from Python fundamentals and classical machine learning, through neural networks and deep learning, to NLP and a complete capstone build. Building the Handwritten Digit Recognizer and Spam Message Classifier end-to-end gave me hands-on confidence in designing, training, evaluating, and delivering real AI systems, complementing my existing full-stack development background.

“The most valuable outcome wasn’t just learning AI concepts — it was learning how to turn them into working, evaluated, and documented software.”

Thank You

Questions & Discussion